How Drishti fits together

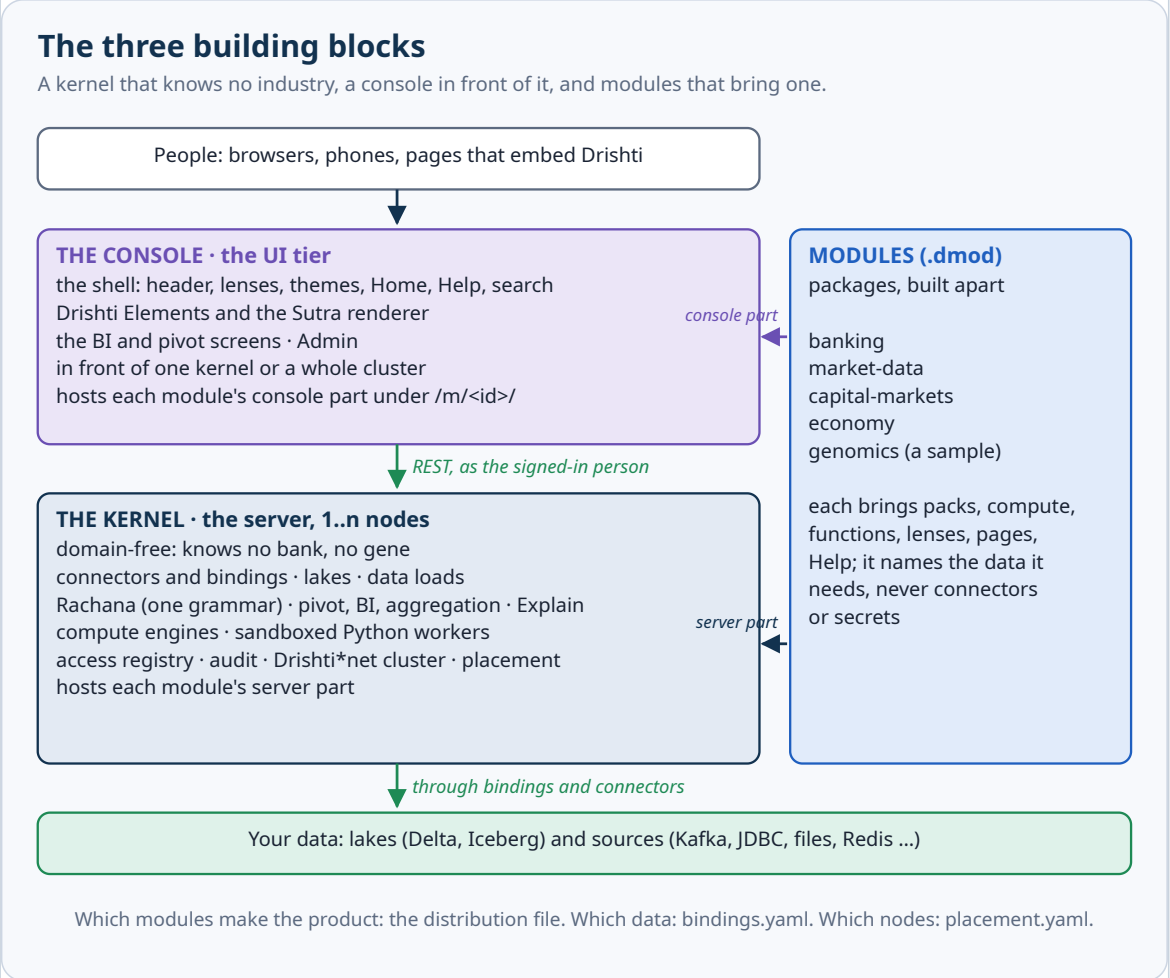
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A short tour for someone new to Drishti: an architect at a bank, a developer, an administrator. One idea per section, one picture per idea, one small real example each. The long references are linked at the end; this page only shows how the pieces connect. Also as a PDF (HOW_IT_FITS.pdf).

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1. The three building blocks



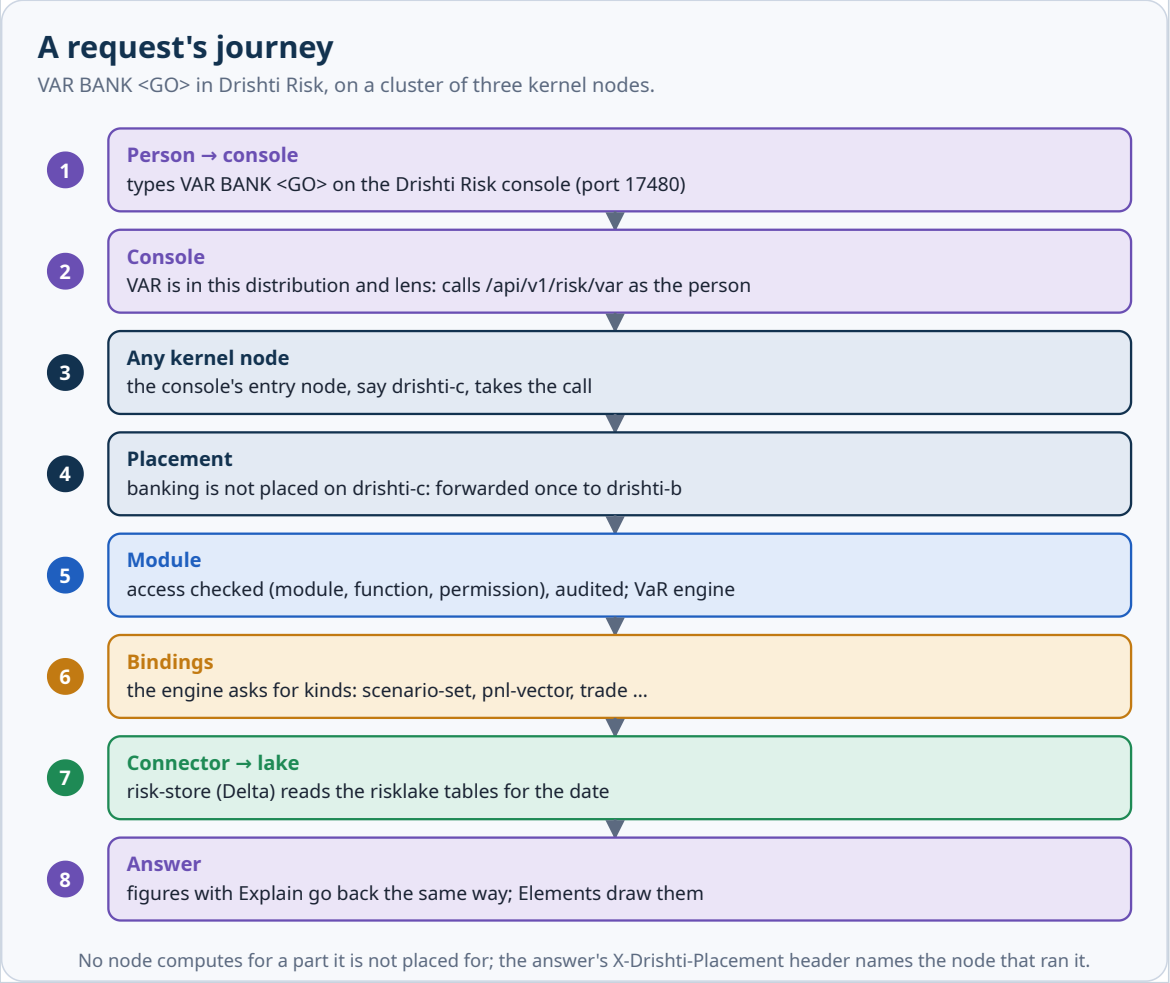
Block	In one line	What it holds
The kernel	the server (Java 21, Spring Boot); one node or a cluster of many	connectors and lakes, data loads, Rachana (the one grammar), the pivot, BI and aggregation engines, Explain, the access registry, audit, the cluster (Drishti*net) and placement; it runs the **server part** of each module
The console	the UI tier, in front of one kernel or a whole cluster	the shell (header, lenses, themes, Home, Help, search), Drishti Elements and the Sutra renderer, the BI and pivot screens, Admin; it shows the **console part** of each module under `/m/<id>/`
Modules	`.dmod` packages, built apart from both	an industry: packs, compute, functions (command words), lenses, pages, Help

The kernel knows no industry: it has no idea what a trade or a gene is. Modules bring that knowledge:

Module	Brings
`banking`	VaR and ES, counterparty exposure and the pre-deal check, sensitivities, FRTB-SA, SIMM; the trading and risk packs
`market-data`	prices, curves, volatilities and FX; needed by banking
`capital-markets` (id `capitalmarkets`)	US equities (filings, facts, indices), funds and ETFs
`economy`	economies, central banks, official statistics, public-sector data and news
`genomics`	a sample third party's module: variants, genes, cohorts; it proves the kernel carries no finance

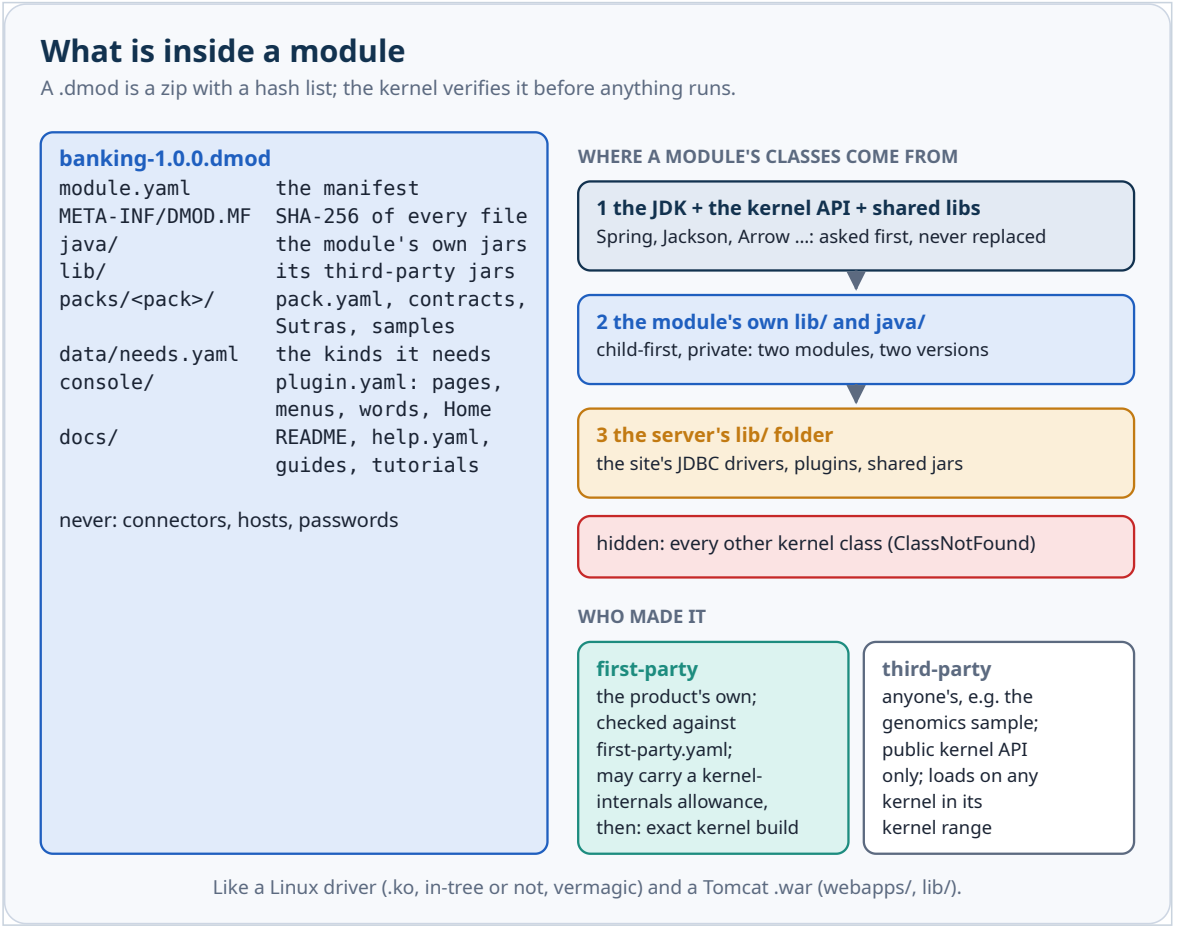
Example. `VAR BANK <GO>` is typed into a console, answered by banking's VaR engine running inside a kernel, from trade-level P&L the bank's own systems loaded into a lake.

2. A request's journey



1. **Person → console.** A risk manager types `VAR BANK <GO>` on the Drishti Risk console.
2. **Console → kernel.** The console knows `VAR` belongs to banking and is offered in this distribution and lens. It calls `/api/v1/risk/var` on its kernel entry node, **as that person** (the kernel decides what the person may see, never the console).
3. **Any node.** Any kernel node of the cluster may take the call; it need not be the one that computes.
4. **Placement.** If banking is not placed on this node, the call is **forwarded once** to a node where it is placed and up. The answer's `X-Drishti-Placement` header names the node that ran it. A node never computes for a part it is not placed for.
5. **Module.** The kernel checks that the module is on, the function is on and the person may use it, writes the audit line, and hands the call to banking's VaR engine.
6. **Bindings.** The engine asks for **kinds** (`scenario-set`, `pnl-vector` or `scenario-pv`, `trade` ...), never for a database. The kernel's bindings say which connector serves each kind.
7. **Connector → lake.** The connector (`risk-store`, a Delta connector) reads that day's tables of the lake (`risklake`).
8. **Answer.** The figures, with how each was calculated (Explain), go back the same way; the console draws them with Drishti Elements.

3. Modules



What a `.dmod` is. A zip with a fixed layout that anyone can produce:

Entry	Holds
`module.yaml`	the manifest: id, version, origin, the kernel range it needs, the modules it requires, the kinds it provides, its functions, engines, packs, API prefix, console plugin
`META-INF/DMOD.MF`	the SHA-256 of every file; one changed byte refuses the whole module
`java/` and `lib/`	the module's own jars, and the third-party jars it carries (listed under `java.lib`)
`packs/<pack>/`	its packs: `pack.yaml`, the contracts of its kinds, Sutras (screens in YAML), samples, guides
`data/needs.yaml`	the kinds it reads, which are required and which optional
`console/plugin.yaml`	its console part: pages, menus, command words, Home cards
`docs/`	its README, Help sections, guides and tutorials

What a `.dmod` never holds: a connector, a host name, a URL of a data source, a user name or a password. The packer refuses them.

Checked before anything runs. The kernel verifies the hash list and the manifest and runs fifteen conformance checks (C1–C15: no kernel internals, declared jars only, contracts for its kinds, nothing leaked after unloading, and the rest). A refused module is absent, never half loaded.

First-party and third-party.

- A **first-party** module is the product's own (banking, market-data, capital-markets, economy). Its claim is **verified, never trusted**: the server jar carries `first-party.yaml`, the id and hash-list hash of each, and only a module that matches counts as first-party.
- Only a first-party module may carry a **kernel-internals allowance** (`kernel-internals:` in its manifest): a short list of kernel packages it may still reach, which only ever shrinks. Banking has one. A module with an allowance loads only on the **exact kernel build** it was packed for (*built for kernel X, this is Y: install the matching module*).
- A **third-party** module (the genomics sample, or a vendor's) uses the public kernel API only and loads on any kernel inside its `kernel` range.

Where a module's classes come from, in this order:

1. **The JDK, the kernel API and the shared libraries** (Spring, Jackson, SLF4J, Arrow …): always asked first, never replaced by a module.
2. **The module's own jars** (`java/`, `lib/`): private to it, so two modules may carry two versions of one library.
3. **The server's `lib/` folder** (`drishti.lib.path`): jars the site adds for everyone (a JDBC driver, a connector plugin, a company library).

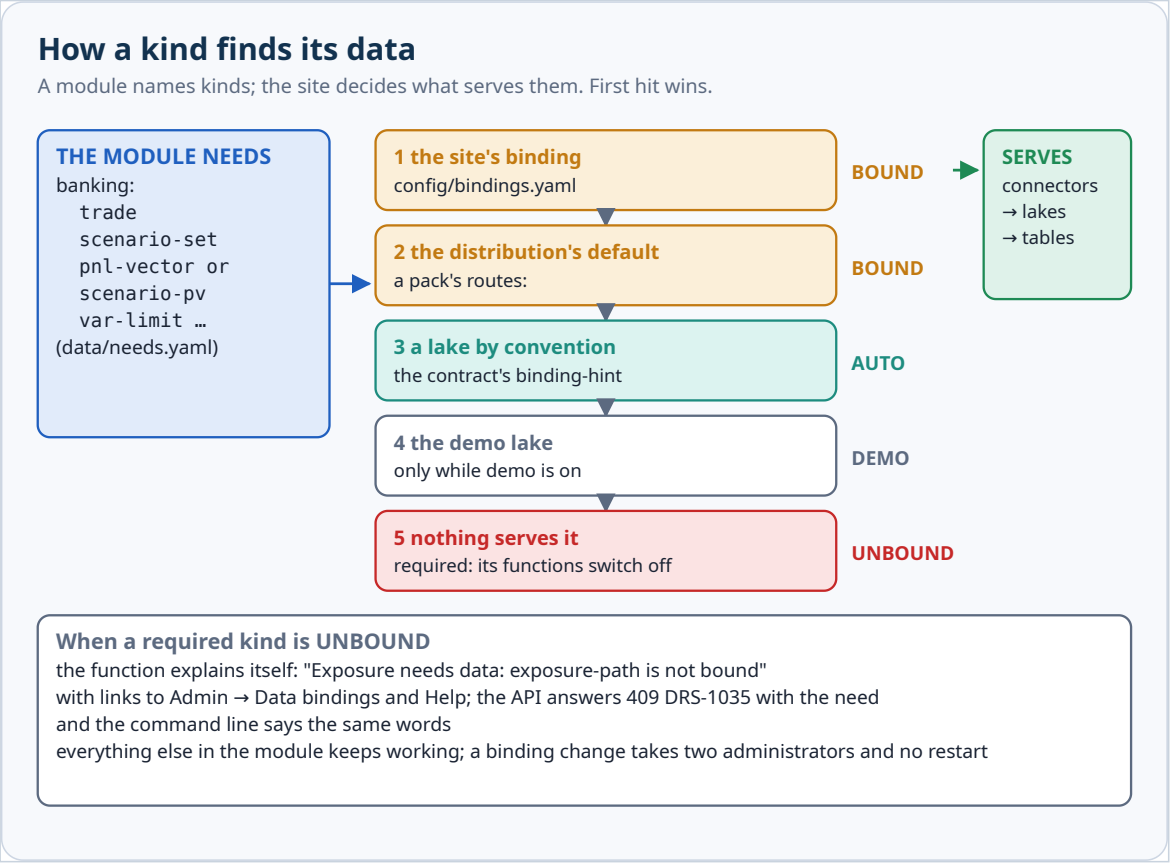
Every other kernel class is hidden from a module, except a verified first-party module's allowance.

Two analogies. A module is to the kernel what a **Linux driver** is to Linux: built apart, version-checked against the kernel (`vermagic` here is the kernel range, or the exact build for an allowance), in-tree (first-party) or out-of-tree (third-party). And the server's folders work like **Tomcat**'s: `modules/` is `webapps/` (drop a package in, it deploys), `lib/` is Tomcat's `lib/` (shared jars).

Example. `banking-1.0.0.dmod` says `origin: first-party`, `requires: market-data`, function `VAR` at `/api/v1/risk/var`, engines `var`, `exposure`, `frtb-sa`, `simm` and the packs `trading`, `market-risk`, `counterparty-risk`. The genomics sample says `origin: third-party` and brings its own VCF reader, kinds `variant`, `sample`, `gene`, `cohort` and a lollipop visual.

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4. Data: bindings



A module says **what** data it needs (kinds); the site decides **where** it comes from. That decision is a **binding**, one per kind, shared by every module that needs the kind, in `config/bindings.yaml`:

```
format: 1
bindings:
  trade:
    read: [trade-endur, trade-calypso]        # connectors (config/connectors/<name>.yaml), in read order
    union: true                               # rows from all of them: one trade store
    live: trade-kafka                         # live changes from a Kafka connector
  scenario-pv:
    read: [risk-store]                        # a Delta connector on the risklake lake
  var-official:
    write: { lake: risklake, table: banking.var-official }   # where a produced kind is written
  pnl-vector:
    demo: true                                # the demo lake's samples, on purpose
```

A binding names connectors and lakes only. Hosts, URLs and credentials stay in the connector files and the secret store, and no module ever sees them.

How a kind finds its data, first hit wins:

#	Source	State
1	the site's binding in `config/bindings.yaml`	`BOUND`
2	the distribution's default (a pack's `routes:`)	`BOUND`
3	a configured lake that holds the contract's `binding-hint` table	`AUTO` (confirm it)
4	the demo lake, only while demo is on	`DEMO`
5	nothing	`UNBOUND`

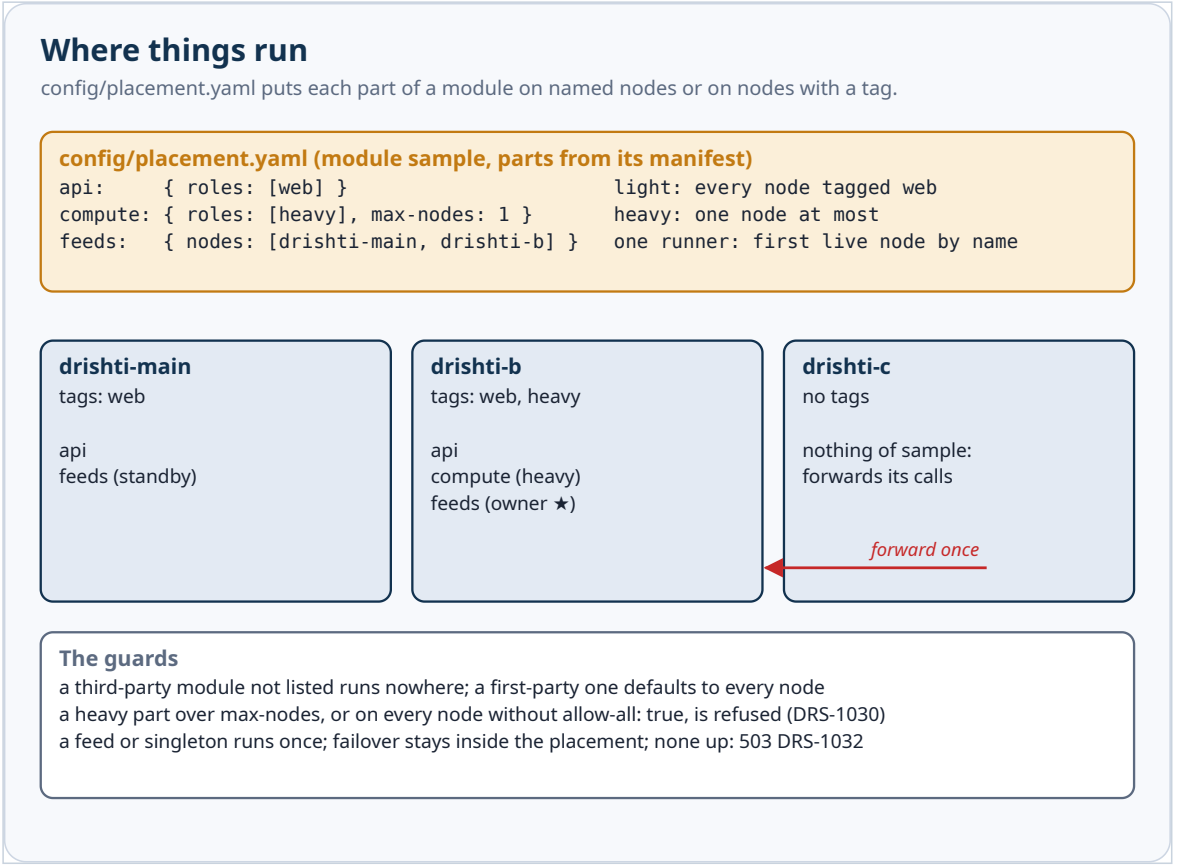
A bound kind with no load inside its contract's cadence is `STALE`; one whose source breaks the contract is `MISMATCH` (treated as unbound); a kind a module computes itself (`position`, `var-result`) is `COMPUTED` and never bound.

Gated functions explain themselves. When a **required** kind is unbound, only the functions that need it switch off, and they say why: the page is *Exposure needs data: exposure-path is not bound* with links to Admin → Data bindings and Help, the API answers `409 DRS-1035` with the need, the command line says the same words. The rest of the module keeps working. An optional kind that is unbound only hides its features (backtesting without `desk-pnl`).

Changing a binding: Admin → Data bindings or the file; one administrator proposes, a second approves; it applies without a restart.

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5. Where things run: placement



On a cluster, `config/placement.yaml` says which nodes run each **part** of a module. The module's manifest declares its parts and their **resource class** (`light`, `heavy` with its memory, `feed`, `singleton`); the site decides where they go, by node name or by node tag (`tags:` in `config/net/node.yaml`). A module that declares no parts has one, `api`, light.

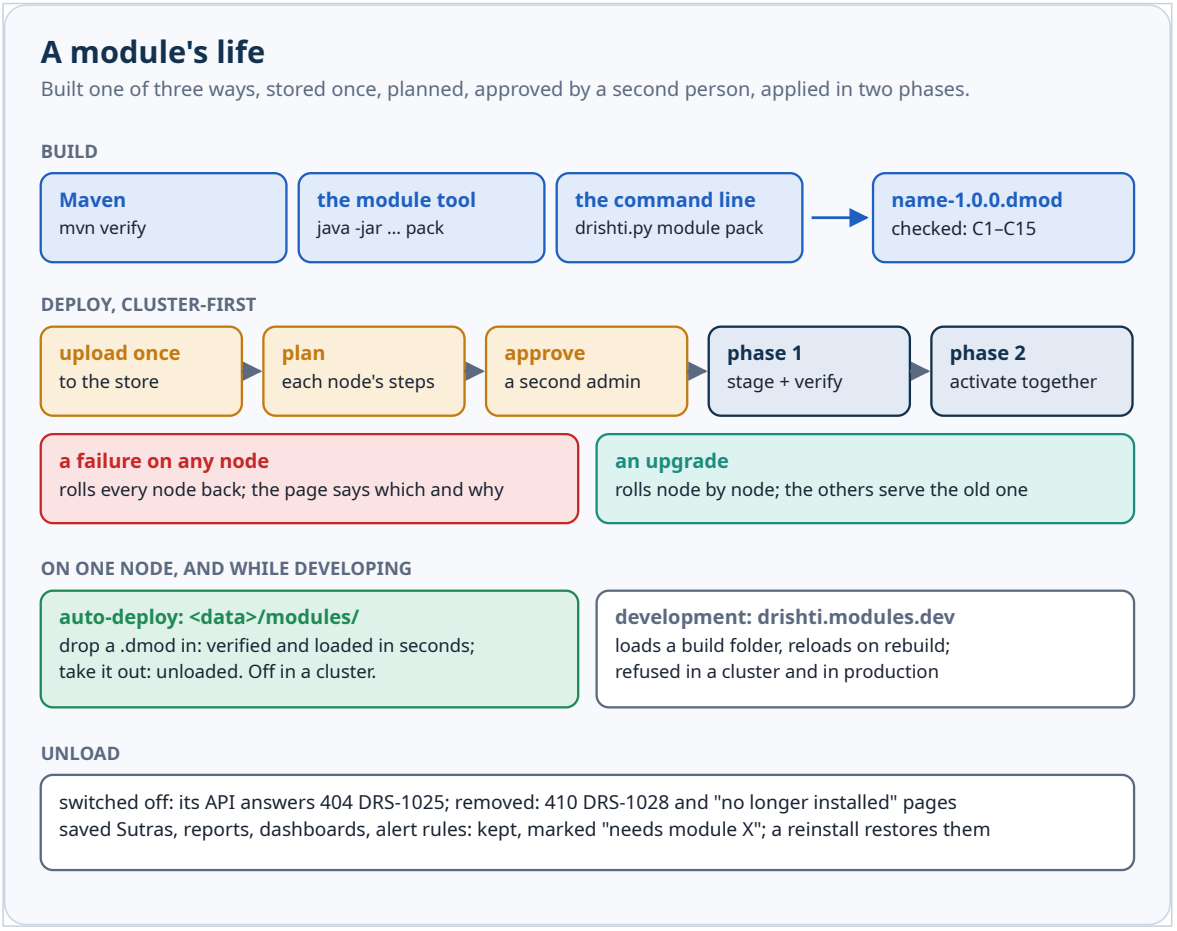
```
format: 1
modules:
  sample:
    parts:
      api:     { roles: [web] }                    # every node tagged web
      compute: { roles: [heavy], max-nodes: 1 }    # heavy: one node at most
      feeds:   { nodes: [drishti-main, drishti-b] } # one runner, elected among these two
```

- **Not listed, not run.** A third-party module or part that the file does not name runs nowhere (*not placed*). A first-party module the file does not name runs on every node, as it always has.
- **Singletons run once.** A `feed` or `singleton` part has exactly one owner: the first live placed node by name. When it stops, the next placed node takes over, never one outside the placement.

- **Heavy work does not spread by accident.** A heavy part over its `max-nodes`, or on every node without `allow-all: true`, is refused (`DRS-1030`).
- **Forwarding.** A node where a part is not placed forwards the call once, for the same person, to a placed node that is up. With none up, the call answers `503 DRS-1032` (*unavailable*, not *absent*: nothing saved is touched).
- **Changing it**: Admin → Cluster → Modules: one administrator plans, a second applies; every member takes it at once, without a restart.

On a single node (no cluster) every part runs on that node: a module the file does not name is placed there by default, and Health notes *single node: placed here*.

6. A module's life



Build, three ways, one packer. The same packer and the same fifteen checks, reached from:

Way	For	Command
Maven	a project made from the archetype `drishti-module-archetype`	`mvn verify` → `target/dmod/<id>-<version>.dmod`
The module tool	Gradle, Bazel, a script, a plain folder	`java -jar drishti-module-tool.jar pack <folder>`, then `check` and `inspect`
The command line	Linux, macOS and Windows with Java and Python only	`python3 tools/drishti.py module pack <folder>`, `module check`, `module inspect`

Deploy, cluster-first.

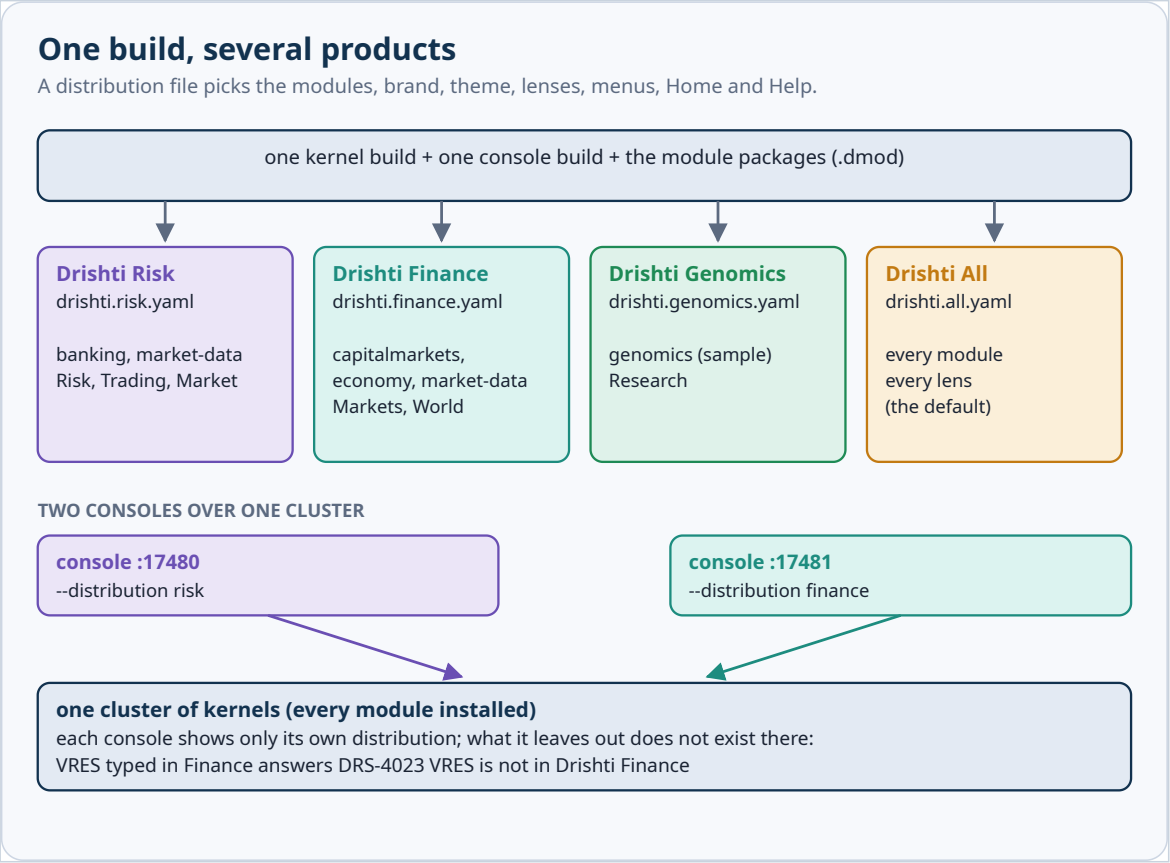
1. **Upload once** to the module store (Admin → Modules → Deploy, or `drishti.py module deploy <file>`). It is checked there once and kept by id and version; a version is never rebuilt with other bytes.
2. **Plan**: what each node does now and after; nodes where nothing is placed receive nothing.
3. **Approve and apply**: a **second administrator** applies the plan (`drishti.py module apply <id>`); the proposer cannot.
4. **Two phases**: every placed node receives the package, checks it itself and stages it; then all activate together. A failure on any node rolls every node back, and the page says which node and why.
5. **Upgrade**: a newer version is staged everywhere first, then switched **node by node** while the others serve the old one.

Auto-deploy on one node. Drop a `.dmod` into `<data folder>/modules/` and within seconds it is verified, installed and loaded; take the file out and the module is removed. It is on for a single node and off in a cluster, where deploys go through plan and apply.

Development folders. With `drishti.modules.dev` a developer's own server loads a module's build folder unpacked and reloads it on every rebuild; Health marks it *development (unverified)*. It is refused in a cluster and under the production profile.

Unload and "needs module X". Switched off, a module's API answers `404 DRS-1025`; removed, `410 DRS-1028` and its pages say it is no longer installed. Saved Sutras, reports, dashboards, baskets, searches and alert rules that use it are **kept**, marked *needs module X*; reinstalling restores them.

7. Distributions (white label)



A **distribution** is one file that turns the kernel into a product: its modules, brand, theme, lenses, menus, Home and Help. One kernel build and one console build make all of them.

Distribution	For	Modules	Start
Drishti Risk	traders, risk analysts and managers	banking, market-data	`bin/start-all.sh --distribution risk`
Drishti Finance	investment and research analysts, economists	capital-markets, economy, market-data	`bin/start-all.sh --distribution finance`
Drishti Genomics	the sample that proves the kernel is domain-free	genomics	`bin/start-all.sh --distribution genomics`
Drishti All	developers, QA, demos; the default	every module	`bin/start-all.sh`

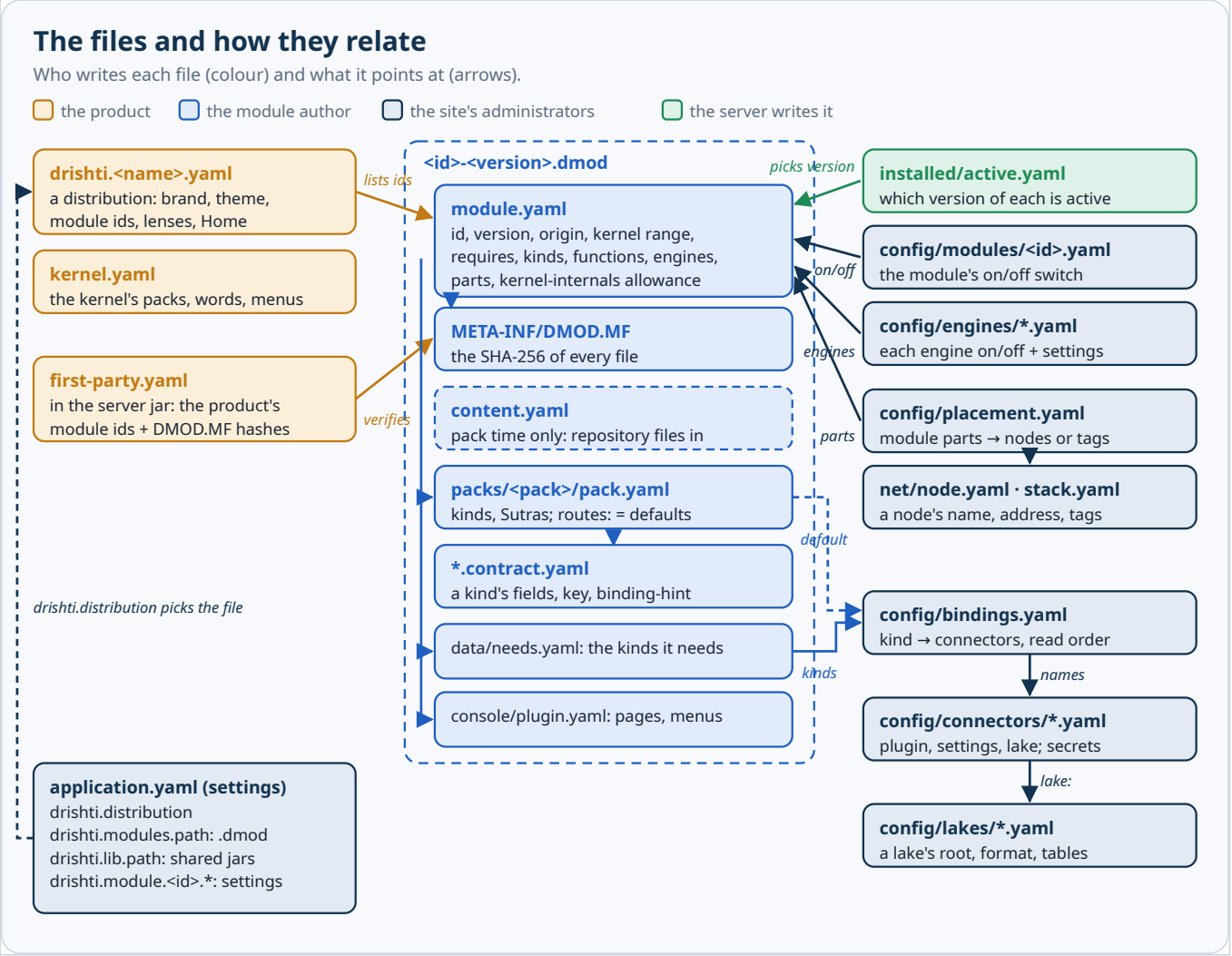
`config/distributions/drishti.risk.yaml`, abridged:

```
format: 1
distribution: drishti.risk
brand:
  product: Drishti Risk
  tagline: VaR, exposure and limits from your own pricing data
theme: { tokens: terminal-blue, mode: dark, density: compact }
modules:
  - { id: banking,     required: true }
  - { id: market-data, required: true }
lenses:
  offered: [risk, trading, market]
  default: risk
home:
  cards: [VAR BANK, VAR DESK RATES, "VRES DESK:RATES", "EXP CPTY:CP-ALDERSHOT"]
docs: { title: Drishti Risk Help }
```

What a distribution leaves out **does not exist in it**: not on the command line, in menus, Home, Help or the catalogue the AI reads. `VRES` typed in Drishti Finance answers *DRS-4023 VRES is not in Drishti Finance*.

- **Choosing one**: `--distribution <name|path>` on `bin/start-all`, `start-server` and `start-console` (`.sh`, `.ps1`, `.cmd`), else `DRISHTI_DISTRIBUTION`, else `drishti.distribution`. On Windows: `bin\start-all.ps1 -Distribution risk`.
- **Two consoles over one cluster**: a console started with its own distribution keeps it whatever the servers run. A stack file's `consoles:` list (`{ name: risk, port: 17480, distribution: risk }`, `{ name: finance, port: 17481, distribution: finance }`) gives Drishti Risk and Drishti Finance side by side over the same kernels.
- **Your own brand**: copy a shipped file to a new `drishti.acme.yaml` beside it in `config/distributions`, change `distribution:`, `brand:`, `theme:`, `modules:`, `lenses:`, `menus:`, `home:` and `docs:`, and start with `--distribution acme`. A distribution names modules; it never contains code, connectors or credentials, and it cannot widen the licence. Admin → White label shows every distribution side by side.

8. The files and how they relate



8.1 Every file, in one table

Who writes it, when it is read, what it decides and where an administrator sees it.

File	Written by	Read	Decides	Admin page
Inside a `.dmod`				
`module.yaml`	the module author	at verification and load; by the distribution reader	id, version, origin, kernel range, required modules, kinds, functions, engines, packs, API prefix, parts, the allowance	Modules (detail), `module inspect`
`META-INF/DMOD.MF`	the packer	at every verification, on every node	the SHA-256 of every file in the package	Modules (verify report)
`content.yaml`	the module author	**at pack time only**, by the packer	which repository files (packs without their connector templates, lenses, tutorials) go into the package	none
`packs/<pack>/pack.yaml`	the module author	when packs load	the pack's kinds and Sutras; its `routes:` are the distribution's default bindings	Packs
`packs/<pack>/contracts/*.contract.yaml`	the module author	when packs load; at binding checks	a kind's fields, key, cadence and `binding-hint`	Data bindings (state)
`data/needs.yaml`	the module author	at load; at every binding resolution	the kinds the module needs, required or optional	Data bindings
`console/plugin.yaml`	the module author	by the console when the module is active	the module's pages, menus, command words, Home cards	Modules
The product				

File	Written by	Read	Decides	Admin page
`first-party.yaml` (inside the server jar)	the product build (`packaging/dist`)	at verification	which modules are the product's own (id + hash-list hash), so which may use an allowance	Health, `module inspect` (origin)
`config/distributions/drishti.<name>.yaml`	the product; a site for its own brand	at start, by the kernel and the console	brand, theme, modules, lenses, menus, Home, Help title	White label (read only)
`config/distributions/kernel.yaml`	the product	at start	the kernel's own packs, words and menus, in every distribution	White label
The site				
`config/bindings.yaml`	the data administrator (four eyes)	at start, then re-read (`drishti.bindings.refresh-seconds`, 30)	which connectors or lake table serve each kind	Data bindings
`config/placement.yaml`	the cluster administrator (plan, then a second applies)	at start; at once on apply, on every member	which nodes run each part of each module	Cluster → Modules
`<data folder>/modules/installed/active.yaml`	**the server** (deploy, auto-deploy, upgrade, rollback)	at start and at each activation	which installed version of each module is active	Modules
`config/modules/<id>.yaml`	the administrator	at start	the module's on/off switch (`drishti.modules.<id>.enabled` wins over it)	Modules (switch off / on)
`config/connectors/*.yaml`	the data administrator; pack templates write first copies	watched; a change applies without a restart	a source: plugin, settings, lake; credentials only as secret references	Connectors
`config/lakes/*.yaml`	the data administrator	with the connectors that use the lake	a lake's format, root and tables	Lakes
`config/engines/*.yaml`	the product's defaults; the administrator	at start; the switch from the Admin page	each compute engine on or off, and its settings (VaR confidence, window …)	Engines
`config/net/node.yaml`	the administrator of each node	watched	this node's name, real address, tags (the `roles:` placement matches)	Cluster → Files
`drishti-stack/stack.yaml`	the operator	by `bin/start-cluster.sh`	the nodes to start, their ports and heaps, the consoles and each one's distribution	`bin/status.sh`
`application.yaml` and settings	the operator	at start	`drishti.distribution`; `drishti.modules.path` (the `.dmod` folder), `drishti.modules.autodeploy`, `drishti.modules.dev`; `drishti.lib.path` (shared jars); `drishti.module.<id>.*` (a module's own settings)	Configuration (Help → Configuration)

8.2 How they point at each other

- A **distribution** lists **module ids** → each id resolves to an installed `.dmod`, and `active.yaml` picks its version.
- `first-party.yaml` verifies the module's origin against its `DMOD.MF`, which hashes every file; only then may the `module.yaml` allowance be used.
- `module.yaml` names its **packs** (whose **contracts** define the kinds), its `data/needs.yaml` and its `console/plugin.yaml`.
- The kinds in `needs.yaml` are mapped by `bindings.yaml` to **connectors** (or, unbound, by a pack's `routes:` default), and each connector names its **lake**.
- `placement.yaml` names the **parts** declared in `module.yaml` and puts them on **nodes** known by their `node.yaml` names and tags (in a local cluster, the **stack file** creates those nodes).
- `config/modules/<id>.yaml` switches the module, and `config/engines/*.yaml` switches each engine it declares.
- `application.yaml` chooses the distribution and the folders everything else is found in.

8.3 One trace: `VAR BANK` in Drishti Risk

Every file the request touches, from start to answer:

#	File	What it contributes
1	`drishti-stack/stack.yaml` or `bin/start-all.sh --distribution risk`	starts the server and a console with the distribution `risk`

#	File	What it contributes
2	`config/distributions/drishti.risk.yaml` (+ `kernel.yaml`)	brand *Drishti Risk*, modules `banking` and `market-data`, the Risk lens, the Home card `VAR BANK`
3	`<data folder>/modules/installed/active.yaml`	banking `1.0.0` is the active version (the full server jar installs the shipped `.dmod` at the first start)
4	`META-INF/DMOD.MF` + `first-party.yaml`	the package is intact and is the product's own, so banking's allowance is honoured on this exact kernel build
5	`config/modules/banking.yaml`	`enabled: true`
6	banking's `module.yaml`	requires `market-data`; function `VAR` at `/api/v1/risk/var`; engine `var`; packs `market-risk` and the others
7	`config/engines/var.yaml`	the VaR engine is on; confidence 0.99, ES 0.975
8	`config/placement.yaml`	banking is first-party: not named, so it runs on every node; named, only where placed (others forward)
9	banking's `console/plugin.yaml`	the console's `VAR` page (route `var`, function `VAR`)
10	banking's `data/needs.yaml`	VaR needs `scenario-set`, `trade` and `pnl-vector` or `scenario-pv`
11	`packs/market-risk/contracts/scenario-pv.contract.yaml`	the kind's key (`SCENARIO_PV_ID`), cadence (daily), `binding-hint` (`risk.scenario-pv`)
12	`config/bindings.yaml`, else `market-risk`'s `routes:`	`scenario-pv` → the connector `risk-store`
13	`config/connectors/risk-store.yaml`	plugin `delta`, its lake
14	`config/lakes/risklake.yaml`	format `delta`, root `<data folder>/delta/risklake`, the tables

The answer goes back through the kernel to the console, which draws it; Explain on any figure shows how it was calculated.

9. Where to read more

To understand	Read
the full design of the kernel, modules, loading, data needs, placement and the distribution file	KERNEL_AND_MODULES.md
each distribution, what is shared, and how to start it	DISTRIBUTIONS.md
building a module: the SDK, the archetype, the three ways to pack, the genomics sample step by step	DEVELOPER_GUIDE.md section 1.6
modules on or off, the `modules/` and `lib/` folders, data bindings, white label, Admin → Modules and Deploy	ADMIN_GUIDE.md section 22 to section 25 (docs/guides/ADMIN_GUIDE.md#25-modules-and-deploy)
several servers as one, placement and deploying across a cluster	CLUSTER_GUIDE.md section 1.9 and 1.10 (docs/guides/CLUSTER_GUIDE.md#110-deploying-a-module)
lakes, connectors and the bindings file in detail	DATA_LAKE_GUIDE.md, Data bindings
installing, starting and checking each part on Windows, Linux and macOS	DEPLOYMENT_AND_QA_GUIDE.md
every setting	CONFIGURATION.md